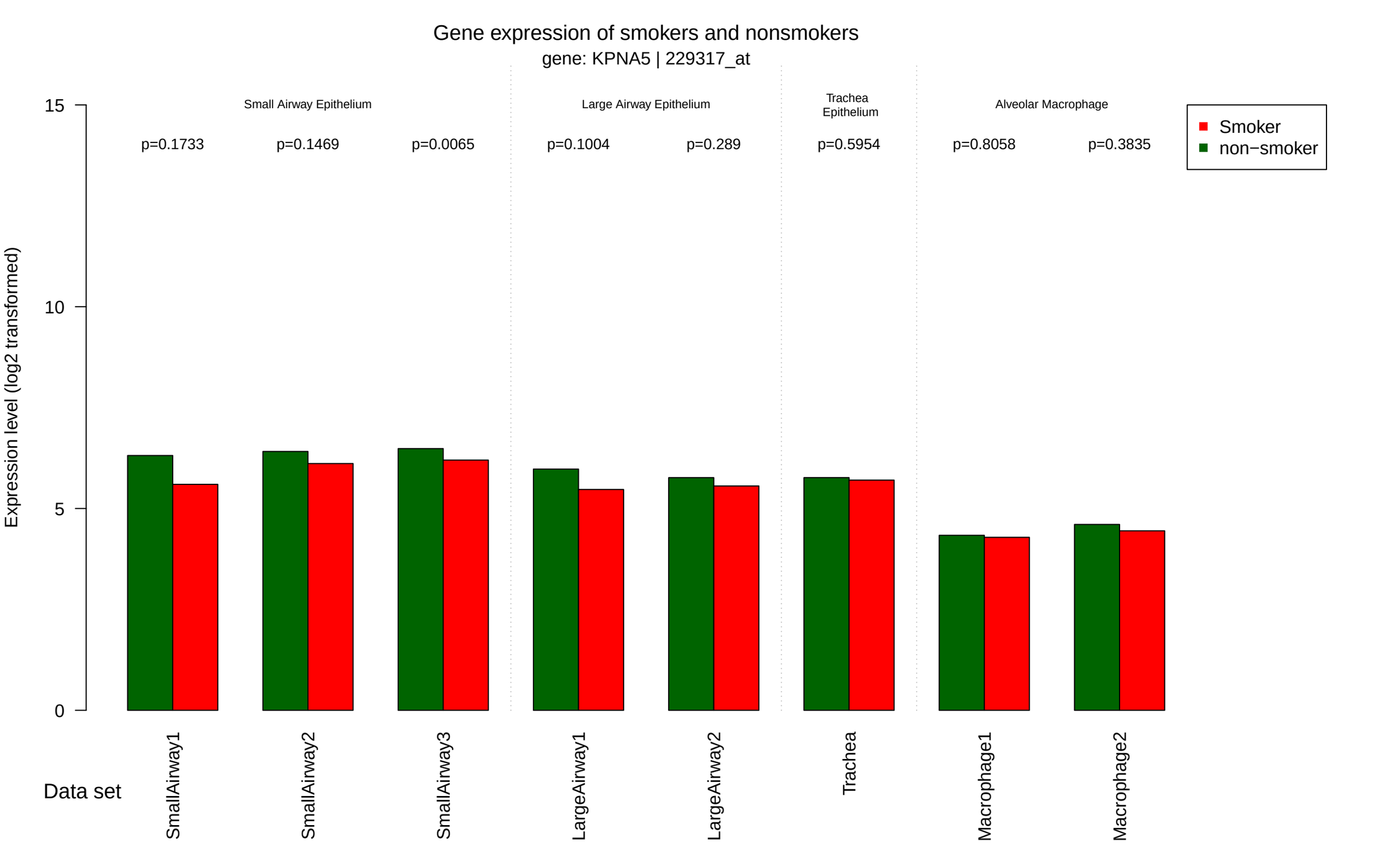
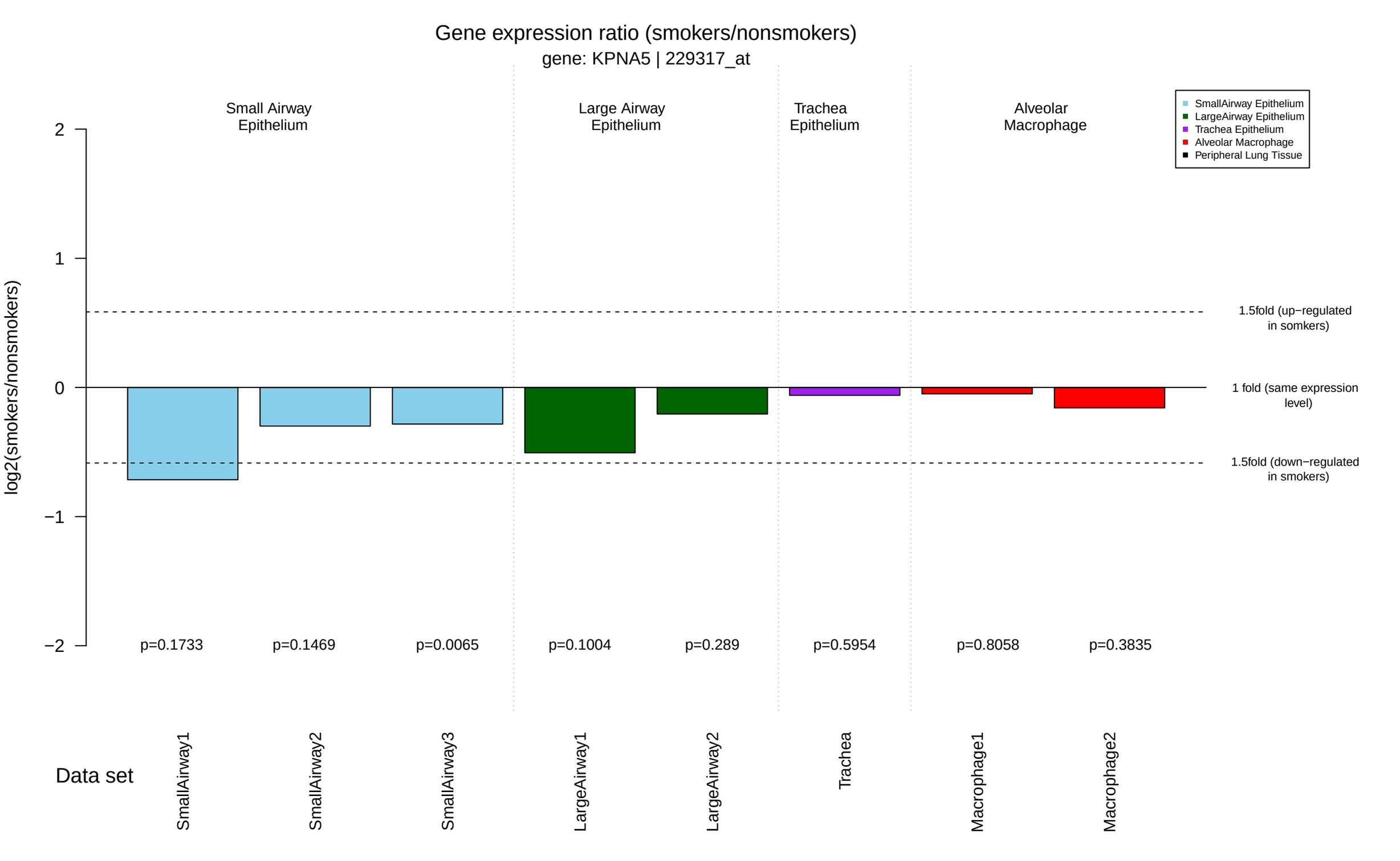
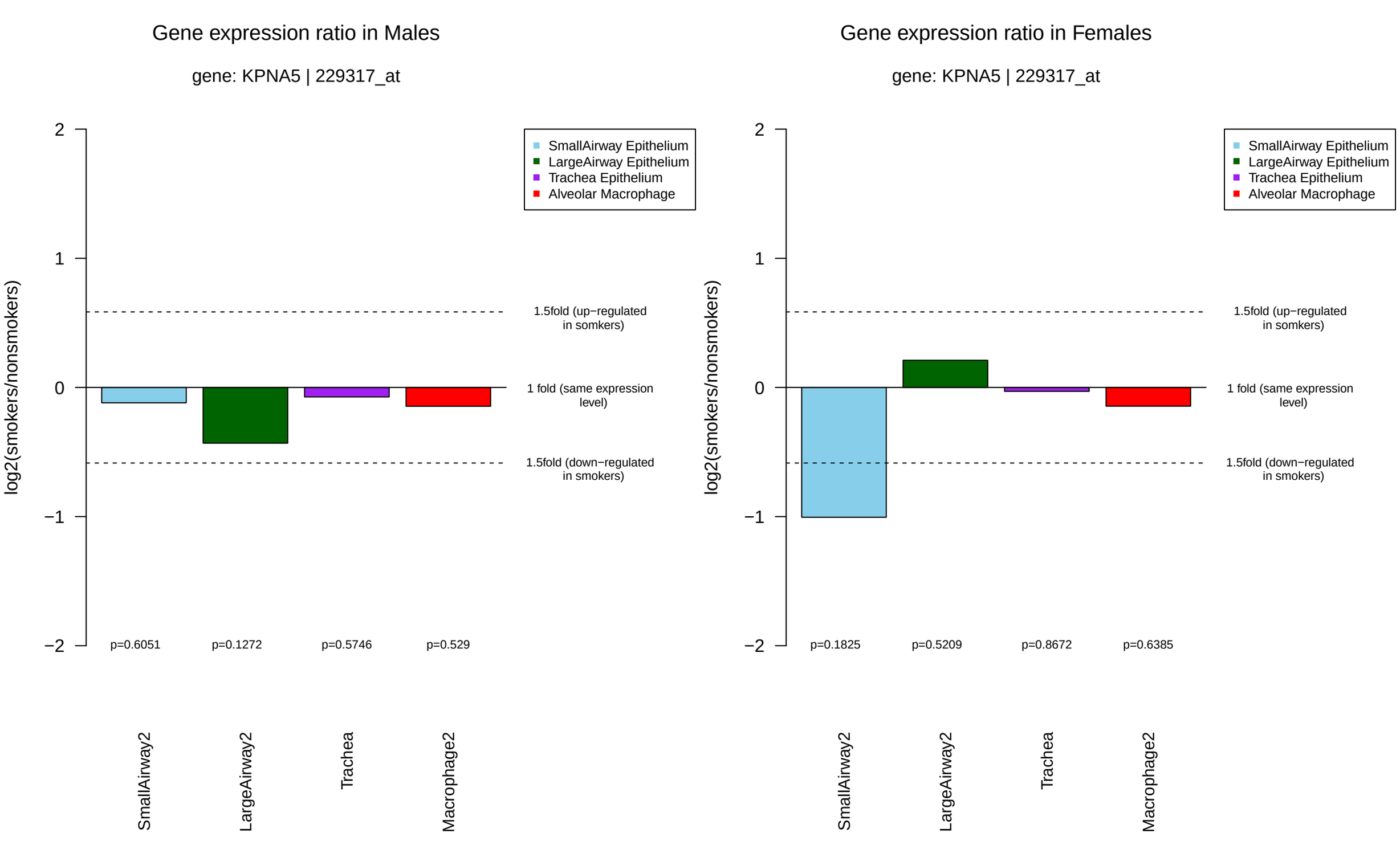
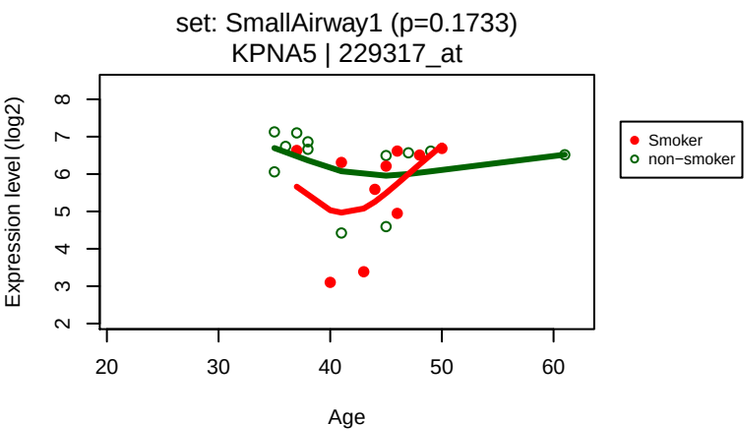
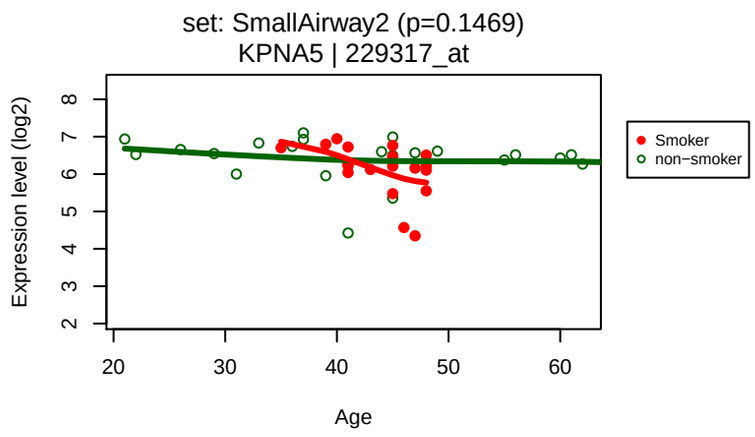
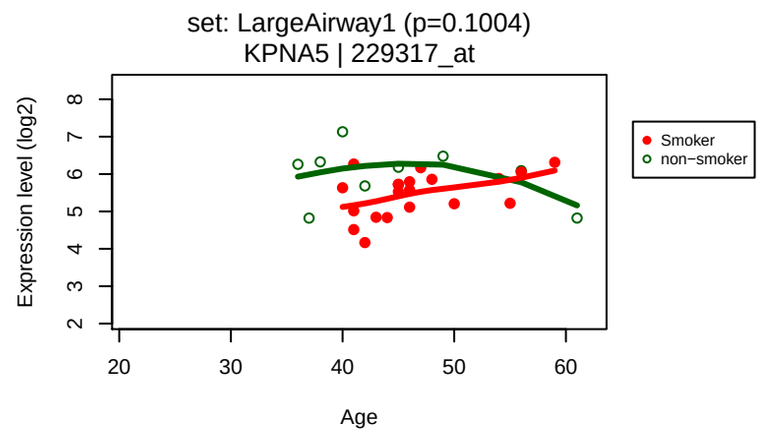

Gene expression of smokers and nonsmokers
gene: KPNA5 | 229317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1733
p=0.1469
p=0.0065
p=0.1004
p=0.289
p=0.5954
p=0.8058
p=0.3835
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

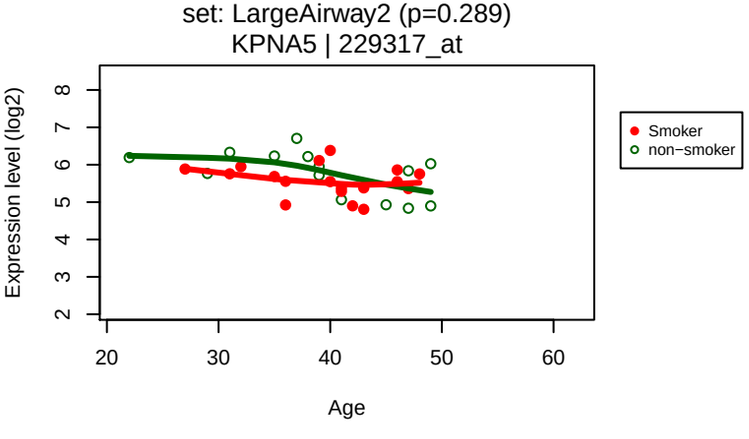

Gene expression ratio (smokers/nonsmokers)
gene: KPNA5 | 229317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1733
p=0.1469
p=0.0065
p=0.1004
p=0.289
p=0.5954
p=0.8058
p=0.3835
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

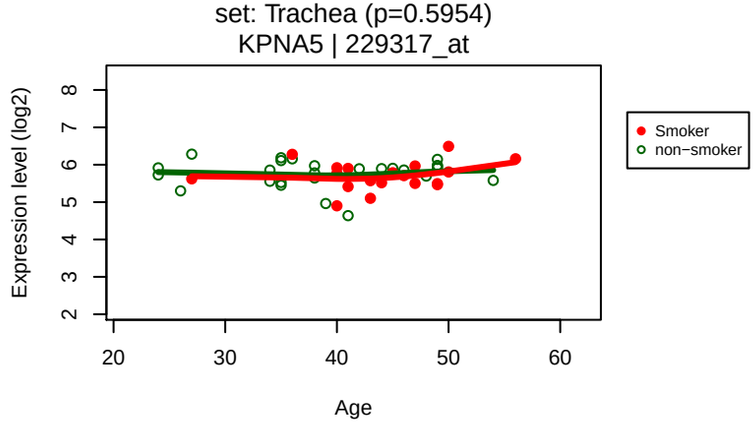

Gene expression ratio in Males
gene: KPNA5 | 229317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6051
p=0.1272
p=0.5746
p=0.529
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA5 | 229317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1825
p=0.5209
p=0.8672
p=0.6385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

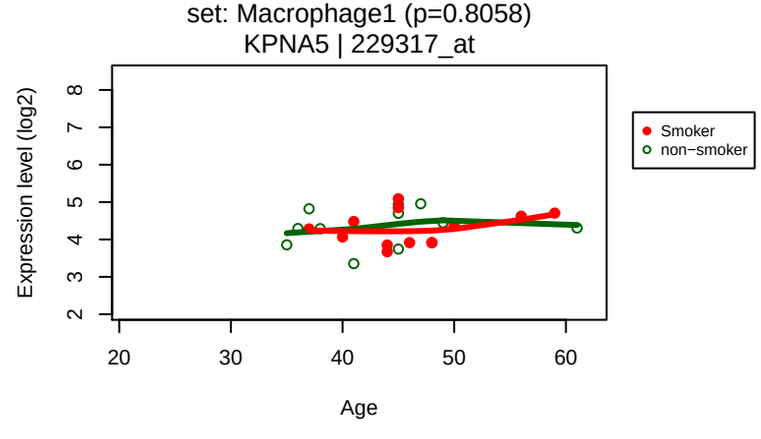

set: SmallAirway1 (p=0.1733)
KPNA5 | 229317_at
Expression level (log2)
Age
Smoker
non−smoker

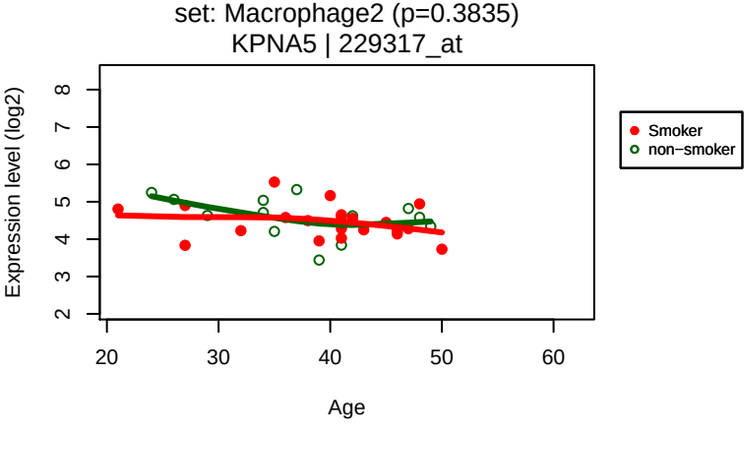

set: SmallAirway2 (p=0.1469)
KPNA5 | 229317_at
Expression level (log2)
Age
Smoker
non−smoker

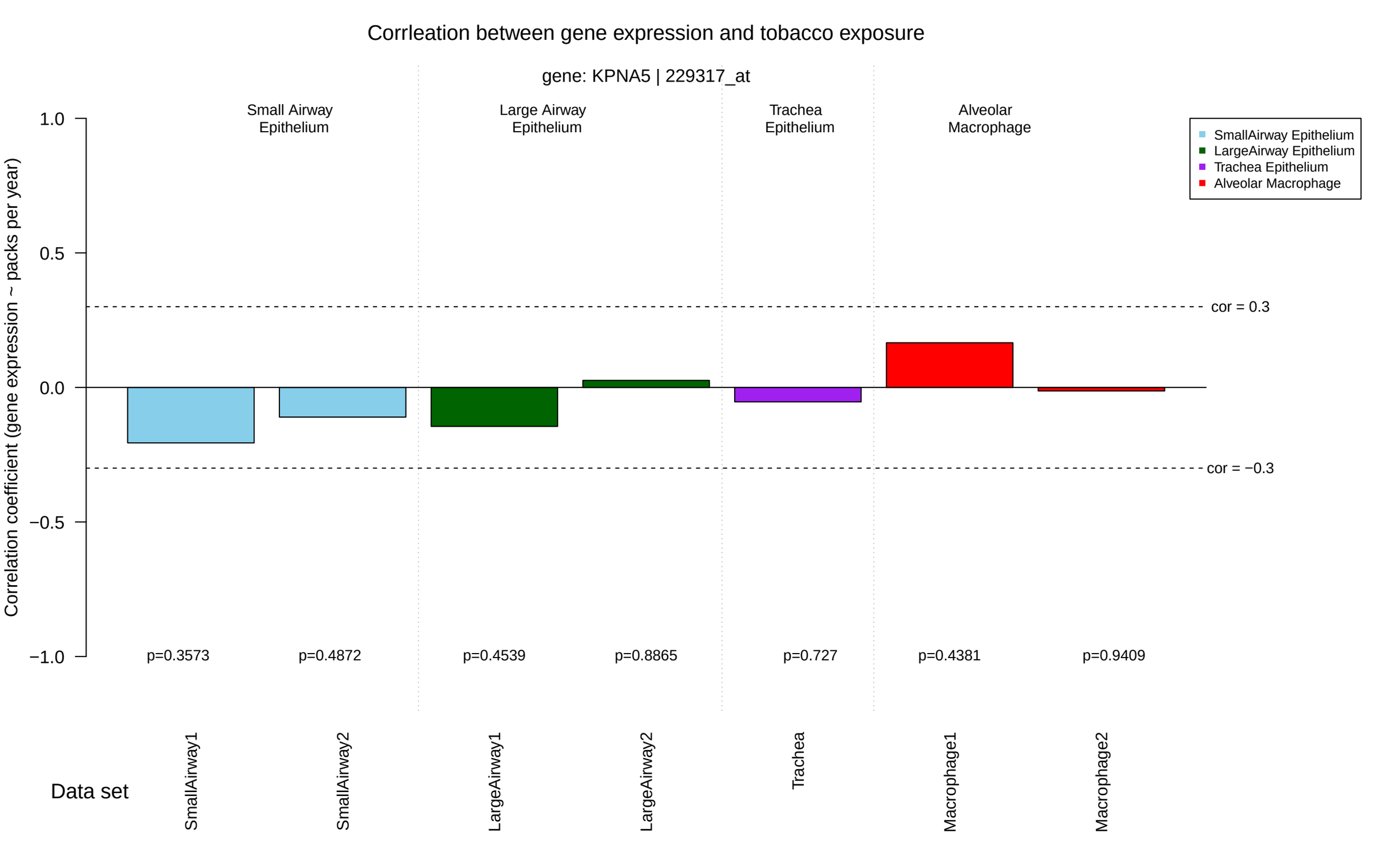

set: LargeAirway1 (p=0.1004)
KPNA5 | 229317_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.289)
KPNA5 | 229317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5954)
KPNA5 | 229317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8058)
KPNA5 | 229317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3835)
KPNA5 | 229317_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA5 | 229317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3573
p=0.4872
p=0.4539
p=0.8865
p=0.727
p=0.4381
p=0.9409
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2